

spiR to Dashboard Visualization Mapping

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Scope: Current SPI dashboard and the local development checkout of `spiR`

Purpose: Keep `spiR` as the single source of truth for reusable SPI visualizations so the dashboard can be replicated for any country, year, and future data release.

Principle

The dashboard should own navigation, controls, layout, loading states, and error states. The `spiR` package should own reusable data retrieval and visualization logic. A dashboard wrapper is acceptable when it adapts Shiny state to an `spiR` function, but it should not reimplement the same chart.

The preferred call pattern is:

```
spiR::spi_plot_function(...)
```

The dashboard must load and verify the development package with:

```
devtools::load_all("../spiR")
```

The external `spiR` repository must not be modified from this dashboard repository.

Official spiR Visualization API

The following functions were verified as exported by the local `spiR` checkout:

spiR function	Signature	Intended output
<code>spi_plot_pillars()</code>	<code>country, pillars, version</code>	Five SPI pillar scores over time for one country
<code>spi_plot_trend()</code>	<code>countries, value_col, version</code>	One SPI series over time for one or more countries
<code>spi_plot_country_vs_region()</code>	<code>country, value_col, version</code>	One country versus its official regional aggregate over time
<code>spi_plot_radar()</code>	<code>country, year, version</code>	Country pillar profile versus its official regional aggregate
<code>spi_plot_regions()</code>	<code>regions, value_col, version</code>	SPI series over time for official regional aggregates
<code>spi_plot_region_pillars()</code>	<code>region, weighted, pillars, version</code>	Official regional pillar trajectories over time
<code>spi_plot_map()</code>	<code>value_col, year, country, zoom, interactive, label, version, resolution</code>	Static or interactive world map

Dashboard Mapping

Visual Order by Window

The table below follows the current dashboard layout. Numbering runs from top to bottom; visualizations on the same row are numbered from right to left. Controls, KPI cards, tables, and explanatory text are excluded because this inventory covers

visualizations only.

No.	Window / tab	Position in window	Visualization	Current source	Target spiR function	Status
1	Overview	Row 1, full width	SPI scores by country map	Flourish embed/live map	<code>spiR::spi_plot_map()</code>	Pending migration
2	Overview	Row 2, full width	Score Distribution	Local <code>hist()</code>	None currently available	Dashboard-specific
3	Overview	Row 3, right	Average SPI by Income Group	Local summary plus <code>spi_plot_horizontal_bars()</code>	None currently available	Dashboard-specific
4	Overview	Row 3, left	Average SPI by Region	Local <code>spi_plot_regions()</code> wrapper	<code>spiR::spi_plot_regions()</code>	Partial
1	Country Profile	Row 1, right	Score Over Time	<code>spi_profile_official_trend()</code>	<code>spiR::spi_plot_trend()</code>	Connected
2	Country Profile	Row 1, left	Pillar Performance radar	<code>spi_profile_official_radar()</code>	<code>spiR::spi_plot_radar()</code>	Connected
1	Trends & Progress	Row 1, full width	Regional history	<code>spi_plot_region_history()</code>	<code>spiR::spi_plot_regions()</code>	Partial

Country Explorer, Compare Countries, Explore by Pillar, and Data & Downloads do not currently contain a rendered visualization in the application. The Country Profile tables are also excluded from this chart inventory.

The visualizations that are available in **spiR** but not currently displayed in the dashboard are `spi_plot_pillars()`, `spi_plot_country_vs_region()`, and `spi_plot_region_pillars()`. The map function `spi_plot_map()` is available as well, but Overview still uses the existing Flourish implementation.

Overview

Dashboard visual	Current implementation	Target spiR function	Status	Required follow-up
SPI scores by country map	Existing Flourish embed/live map through <code>flourish_live_map_ui()</code>	<code>spiR::spi_plot_map()</code>	Pending migration	Replace or offer <code>spi_plot_map()</code> as the package-backed map. Preserve Flourish only if it remains an explicit product requirement.
Score Distribution	<code>hist()</code> in <code>app_server.R</code> using <code>overview_score_distribution()</code>	None currently available	Dashboard-specific	Keep local unless a distribution function is added to spiR .
Average SPI by Region	<code>spi_plot_regions()</code> in <code>R/spi_visualizations.R</code> , currently computes a local summary and draws local horizontal bars	<code>spiR::spi_plot_regions()</code>	Partial	Pass region selections and the correct value column directly to spiR ; remove the local chart implementation

Dashboard visual	Current implementation	Target spiR function	Status	Required follow-up
Average SPI by Income Group	<code>spi_plot_horizontal_bars()</code> using <code>overview_income_group_summary()</code>	None currently available	Dashboard-specific	once the output is equivalent. Keep local unless spiR adds an income-group visualization.
Regional history	<code>spi_plot_region_history()</code> wraps <code>spiR::spi_plot_regions()</code> and adds local layers using internal package objects	<code>spiR::spi_plot_regions()</code>	Partial	Use the public function directly and avoid <code>getFromNamespace()</code> for internal spiR objects. If extra layers are needed, add a supported public option or function to spiR .

Country Explorer

Country Explorer is primarily a table and filter workflow. It has no chart currently rendered in the module.

Dashboard visual	Current implementation	Target spiR function	Status
Pillar view table	Normalized data table	<code>spiR::spi_plot_pillars()</code> is not a table replacement	No chart mapping
Dimension view table	Normalized data table	No direct equivalent	No chart mapping
Indicator view table	Normalized data table	No direct equivalent	No chart mapping

A future Explorer chart could use `spi_plot_pillars()` for a selected country, but the table should remain a separate dashboard interaction unless **spiR** exposes a reusable table-oriented API.

Country Profile

Dashboard visual	Current implementation	Target spiR function	Status
Pillar Performance radar	<code>spi_profile_official_radar()</code> in <code>R/country_profile_visualizations.R</code>	<code>spiR::spi_plot_radar(country, year)</code>	Connected
Score Over Time	<code>spi_profile_official_trend()</code> in <code>R/country_profile_visualizations.R</code>	<code>spiR::spi_plot_trend(countries)</code>	Connected
Pillars over time	Not currently shown as a separate chart	<code>spiR::spi_plot_pillars(country)</code>	Available, not assigned
Country versus region	Not currently shown as a separate chart	<code>spiR::spi_plot_country_vs_region(country, value_col)</code>	Available, not assigned

The Profile wrappers exist only to prioritize the official package function and retain a controlled fallback for offline fixtures. New Profile visualizations should follow the same pattern.

Future Dashboard Tabs

Future visual concept	Official spiR function	Status
Country pillar trajectories	<code>spiR::spi_plot_pillars()</code>	Available, not assigned
Country versus regional trajectory	<code>spiR::spi_plot_country_vs_region()</code>	Available, not assigned
Regional pillar trajectories	<code>spiR::spi_plot_region_pillars()</code>	Available, not assigned
Regional SPI trends	<code>spiR::spi_plot_regions()</code>	Available, partially used
World map	<code>spiR::spi_plot_map()</code>	Available, not assigned; current Overview uses Flourish

Coverage Summary

Already backed by official spiR functions

- Country Profile radar: `spi_plot_radar()`.
- Country Profile score trend: `spi_plot_trend()`.
- Part of the regional history path: `spi_plot_regions()`.

Still implemented locally or with a local rendering layer

- Overview world map: Flourish embed/live implementation.
- Overview score distribution: base R histogram.
- Overview regional average bars: local aggregation and bar renderer.
- Overview income-group bars: local aggregation and bar renderer.
- Additional layers in regional history: local code accesses internal **spiR** objects.

Official functions available but not yet assigned

- `spi_plot_pillars()`.
- `spi_plot_country_vs_region()`.
- `spi_plot_region_pillars()`.
- `spi_plot_map()`.

Replication Roadmap

- Keep **spiR** function calls behind small dashboard wrappers that accept Shiny selections and return the plot object.
- Migrate regional history away from internal namespace access and local reconstruction.
- Evaluate `spi_plot_map()` against the current Flourish map and choose one supported source of truth.
- Add `spi_plot_pillars()` and `spi_plot_country_vs_region()` to the Country Profile when the UX requires them.
- Add tests that verify each mapped dashboard output calls the official **spiR** function when the package is available.
- Keep local dashboard-only charts only where there is no corresponding **spiR** function, and record those gaps here.
- When new visualization functions are released in the official CRAN version of **spiR**, update this mapping and replace temporary fallbacks where the contracts are compatible.

Verification Commands

From the dashboard repository root:

```
devtools::load_all("../spiR")

getNamespaceExports("spiR")[grep(
```

```
  "plot|radar", getNamespaceExports("spiR"), ignore.case = TRUE
)]

formals(spiR::spi_plot_radar)
formals(spiR::spi_plot_trend)
```

The mapping should be reviewed whenever the `spiR` API changes or a new dashboard visualization is added.